

Prajwal Kulkarni

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EDUCATION

KLS Gogte Institute of Technology

BE in Electronics and Communication | GPA: 8.98

Belagavi, Karnataka

2022 – 2026

Govindram Seksaria Pre-University College

Score: 92.83%

Belagavi, Karnataka

2020 – 2022

EXPERIENCE

Tech Fortune Technologies

Data Science Intern

Jan 2026 – May 2026

Bengaluru

- Built a Dockerized vehicle tracking REST API using FastAPI and YOLOv8n, maintaining persistent object identities across occlusions (up to **5** missed frames) by implementing SORT with a Kalman Filter.
- Accelerated real-time cloud processing speed on Hugging Face Spaces by bypassing YOLO detection on **50%** of frames via Kalman predictions, utilizing FP16 inference, and encoding browser-ready WebM streams.
- Conducted workshops and classes on Machine and Deep Learning for **100+** BCA/BE/Diploma students.

Department of Neurosurgery, AIIMS

Research Intern

May 2025 – Sept 2025

New Delhi

- Architected a multi-agent Llama AI and Visual RAG system to automate surgical suturing evaluation. Curated a **514-image** dataset, achieving an RMSE of **2.2/10** against expert surgeon benchmarks.
- Deployed a Python/PyQt5 computer vision application with multi-threaded OpenCV live feeds for real-time craniotomy scoring. Integrated Grad-CAM pipeline to visually highlight spatial drilling performance.
- Built a synchronized **3** camera hardware system using C-programmed ESP32S3 microcontrollers and OpenCV. Recorded bio-signal data for endoscopic hand-eye coordination research.

PROJECTS

Medical Model Calibration [Code](#)

- Engineered an end-to-end deep learning framework with **180** experiments across **5** medical datasets containing **26,957** images to analyze effect of train-time regularization and post-hoc calibration techniques.
- Proved standard post-hoc methods worsen reliability in **78.3%** of cases due to overfitting, and instead achieved a peak **97.86%** accuracy with low calibration error by using native Train-Time Regularization.
- *Technologies: Python, Pytorch, Torchvision, OpenCV, Scikit-Learn*

Real-Time Industrial Posture Analysis System [Code](#)

- Engineered an FPS agnostic data pipeline to extract **6** channel kinematic tensors (position + velocity) from the CarDA dataset, training an **SE-STGCN** model to achieve **94.6%** accuracy on complex 3D skeletal sequences.
- Deployed a decoupled PyQt6 edge-client utilizing binary WebSockets and OneEuroFilter signal smoothing, delivering viewpoint invariant, EAWS based safety alerts with **<10ms** end-to-end latency.
- *Technologies: Python, PyTorch, MediaPipe, FastAPI, PyQt6, WebSockets*

TECHNICAL SKILLS

Programming Languages: Python, C/C++

AI/ML: PyTorch, Hugging Face Transformers, OpenCV, Scikit-Learn, Pandas

Development: Linux, Git/GitHub, Docker, FastAPI

ACHIEVEMENTS & CERTIFICATIONS

Graphics and Vision Summer School 2024 IIT Delhi

May 2024 - June 2024

Deep Learning - IIT Ropar (Top 1% | Elite + Gold) NPTEL

Machine Learning Specialization Coursera